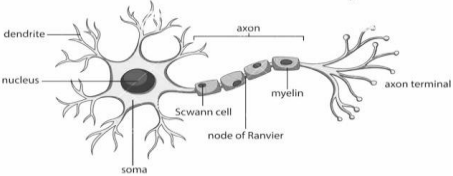
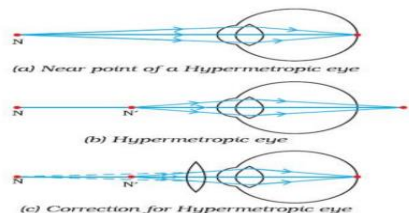
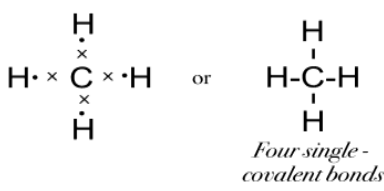
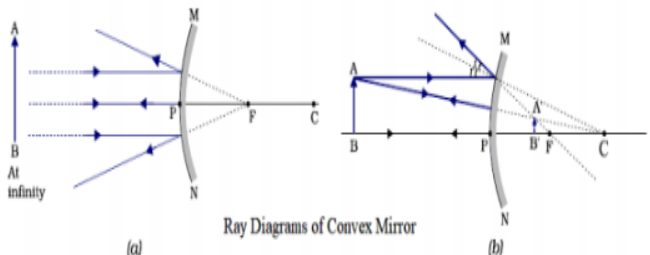


Marking Scheme Pre-board- I Class-X Science (Subject Code – 086)		
Q. No	Answers	Marks
Section–A		
1	c) Loss of electrons	1
2	b) Potassium hydroxide (KOH)	1
3	b) Baking soda	1
4	d) 3/2	1
5	b) Turned green and a coating was formed on the nail	1
6	b) Ethanol	1
7	d) By breaking down the nutrients of bread and then absorbing them	1
8	c) Hormones	1
9	b) Involuntary	1
10	a) Insulin	1
11	c) To protect ovules and develop into fruit	1
12	a) Phenotype	1
13	a) It converges light rays.	1
14	d) Cornea	1
15	d) 2 Ω	1
16	c) R	1
17	C. A is true but R is false.	1
18	A. Both A and R are true, and R is the correct explanation of A.	1
19	D. A is false but R is true	1
20	A. Both A and R are true, and R is the correct explanation of A.	1
Section-B		
Question No.21 to 26 are very short answer questions		
21	<i>Exothermic Reaction:</i> A reaction that releases heat. Example: Combustion of methane :$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$ <i>Endothermic Reaction:</i> A reaction that absorbs heat. Example: Decomposition of calcium carbonate: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ OR A precipitation reaction is a chemical reaction in which two solutions react to form an insoluble solid (precipitate). Example: $\text{AgNO}_3 (\text{aq}) + \text{NaCl} (\text{aq}) \rightarrow \text{AgCl} (\text{s}) + \text{NaNO}_3 (\text{aq})$. Here, AgCl is the precipitate.	2
22	Amphoteric oxides are oxides that behave as both acidic and basic oxides. They can neutralize both acids and bases. They undergo neutralization reaction to form water and salt when reacting with acid and form complex salts and water when reacting with base. Examples – Aluminium oxide (Al_2O_3) and Zinc Oxide (ZnO)	2
23	The process of breaking down food materials within the cell to produce energy and then trapping that energy for ATP production. Total ATP produced = ATP per glucose $\times$ Number of glucose molecules	2

	$= 38 \text{ ATP/glucose} \times 5 \text{ glucose} = 190 \text{ ATP.}$ OR Water lost in a week = Rate $\times$ Hours per day $\times$ Days in a week $= 3 \text{ mL/hour} \times 24 \text{ hours/day} \times 7 \text{ days}$ $= 3 \times 24 \times 7 = 504 \text{ mL.}$ Plants require carbon dioxide from the atmosphere, water from groundwater and solar energy from the sun.	
24	Phenotypic Ratio: 100% Tall plants. Genotypic Ratio: 100% Heterozygous (Tt). In the F1 generation, all the plants will be tall because the tall allele (T) is dominant over the short allele (t).	2
25	Stars twinkle because their light passes through multiple layers of the atmosphere with varying refractive indices, causing the light to bend in different directions, making the star's brightness appear to fluctuate. Planets do not twinkle because they are closer to Earth and appear as extended sources of light. The light from different parts of the planet's image averages out the atmospheric variations, so they do not twinkle.	2
26	The resistance R of a wire is directly proportional to its length l (i.e., $R \propto l$). If the length of the wire is doubled, the resistance will also double. Therefore, if the original resistance was R, it will become 2R when the length is doubled.	2
Section-C		
Question No. 27 to 33 are short answer questions		
27	On a pH scale, any solution having pH value less than 7 is considered acidic and that with a pH value of more than 7 is basic. Hence, according to the given pH values, solution 'A' is acidic in nature, while solution 'B' is basic. Also, since 'A' is acidic, it has a greater concentration of hydrogen ions.	3
28	Calcium starts floating because the bubbles of hydrogen gas formed stick to the surface of metal. $\text{Ca(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$ Magnesium reacts with hot water and starts floating due to the bubbles of hydrogen gas sticking to its surface.	3
29	 A neuron consists of three main parts: the cell body, dendrites, and axon. Cell Body: Contains the nucleus and organelles, and is responsible for the metabolic activities of the neuron. It integrates incoming signals. Dendrites: These are branched extensions that receive signals from other neurons and convey them to the cell body. Axon: A long, thin fiber that transmits nerve	3

	impulses away from the cell body to other neurons or muscles. The axon is often covered by a myelin sheath, which speeds up the conduction of nerve impulses. OR Cerebrum: The largest part of the brain, responsible for higher functions such as thought, learning, memory, and voluntary muscle movements. It is divided into two hemispheres and further into lobes that control different sensory and motor functions. Cerebellum: Located at the back of the brain, it coordinates muscle movements, maintains posture, and balance. It is essential for fine motor skills. Brain stem: Comprising the midbrain, pons, and medulla oblongata, it controls basic life functions such as breathing, heart rate, and blood pressure. The brainstem also serves as a communication pathway between the brain and the spinal cord.	
30	Genotype: The genetic makeup of an organism, represented by the alleles it possesses (e.g., TT, Tt, or tt). Phenotype: The observable characteristics or traits of an organism resulting from the interaction of its genotype with the environment (e.g., tall or short plants). For instance, a plant with the genotype Tt will have the phenotype of being tall.	3
31	In hypermetropia, the eye cannot converge the light rays coming from nearby objects, and the image is formed behind the retina, making nearby objects appear blurry. Hypermetropia can be corrected by using a convex lens. 	3
32	 With the help of demonstration I can help him. This rule is used to find the direction of the magnetic field produced by the current-carrying conductor around it. Statement: If a current-carrying conductor is imagined to be held in the right hand such that the thumb points in the direction of the current, then the curled fingers of the hand indicate the direction of magnetic field. OR Fleming's left hand rule, Statement: Stretch the left hand such that the thumb, first finger and the central finger are mutually perpendicular to each other	3
33	a) 10% 10% Producer → Primary consumer → Secondary consumer 25,000 J 2500 J 250 J b) Aquariums are artificially built ecosystems which generally do not contain every aspect of natural things like decomposers are not present in it. c) According to 10% law of energy transfer only 10% of energy passes from one trophic level to the next. Thus the amount of energy decreases with successive trophic	1+1+ 1=3

	levels.	
Section-D		
Question No.34 to 36 are long answers questions.		
34	a) Covalent bonds are formed when two atoms share one or more pairs of electrons to attain a stable electron configuration (usually that of a noble gas). These bonds typically occur between non-metals. Example: In a water molecule (H₂O), each hydrogen atom shares one electron with oxygen, forming covalent bonds. b) Covalent compounds do not have free ions or electrons to carry an electric charge, which is why they are generally poor conductors of electricity. The electrons in covalent compounds are localized between atoms, not free to move as in ionic compounds. c) Single bond: In a single bond, one pair of electrons is shared between two atoms. Example: H-H (Hydrogen molecule). Double bond: In a double bond, two pairs of electrons are shared between two atoms. Example: O=O (Oxygen molecule). Triple bond: In a triple bond, three pairs of electrons are shared between two atoms. Example: N≡N (Nitrogen molecule).  <i>Four single - covalent bonds</i> d) OR a) A homologous series is a group of organic compounds that have the same functional group, similar chemical properties, and successive members differ by a CH₂ group. All members follow a general formula, e.g., for alkanes, the general formula is C_nH_{2n+2}. Eg: The alkane series includes methane (CH₄), ethane (C₂H₆), propane (C₃H₈), etc. b) Carbon has the unique ability to form long chains and rings of atoms by bonding with other carbon atoms. This property is known as catenation. Carbon-carbon bonds are strong and stable, allowing for the formation of a variety of compounds. Carbon has four valence electrons, meaning it can form four covalent bonds with other atoms. This property allows carbon to bond with hydrogen, oxygen,	5

	nitrogen, sulfur, and other elements to form numerous compounds. c) The combustion of ethane (C_2H_6) in the presence of oxygen produces carbon dioxide and water. $2 C_2H_6 + 7 O_2 \rightarrow 4 CO_2 + 6 H_2O$ d) Saturated hydrocarbons: These are hydrocarbons in which carbon atoms are bonded to other atoms by single bonds only. They are less reactive. Example: Alkanes (methane, ethane). Unsaturated hydrocarbons: These are hydrocarbons that contain one or more double or triple bonds between carbon atoms. They are more reactive. Example: Alkenes (ethylene), Alkynes (acetylene).	
35	a) The male reproductive part is called the stamen, and the female reproductive part is called the pistil. b) Pollination is the transfer of pollen grains from the anther (male part) to the stigma (female part). Pollination can occur through wind, insects, water, or animals. c) Fertilization in plants occurs when the male gamete (pollen) fuses with the female gamete (ovule) inside the ovary to form a zygote. d) After fertilization, the ovary develops into a fruit, and the ovule becomes the seed containing the embryo. OR a) C – Urethra D – Vas deferens b) Testes produce male sex hormone testosterone. Hormone Testosterone brings about the development of secondary sexual characters during puberty in boys like growth of facial hair, deepening of voice, build up of muscle mass and also regulates formation of sperms. c) Seminal vesicles B release its contents into the ejaculatory duct during ejaculation. d) Urethra C carries sperms from the vas deferens through the penis.	1+2+ 1+1= 5
36	 Ray Diagrams of Convex Mirror (a) (b)	2+3= 5

	Focal length (f) = +20 cm Image distance (v) = +30 cm $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ or $\frac{1}{u} = \frac{1}{v} - \frac{1}{f}$ or $\frac{1}{u} = \frac{1}{30} - \frac{1}{20}$ $\frac{1}{u} = \frac{2-3}{60} = -\frac{1}{60}$ $u = -60$ cm The object should be placed 60 cm in front of the lens. OR Characteristics of the image formed by a plane mirror are (i) image distance is same as that of object distance (ii) image formed is virtual and erect (iii) image formed is of the same size as that of the object (iv) image formed is laterally inverted Focal length of convex mirror (f) = +15 cm, Object distance (u) = – 10 cm $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $\frac{1}{v} = \frac{1}{15} - \frac{1}{-10} = \frac{2+3}{30}$ $v = \frac{5}{30} = 6\text{cm}$ $\text{Magnification} = \frac{-v}{u} = \frac{-6}{-10} = 0.6$ The image is located at a distance of 6 cm from the mirror on the other side of the mirror. Magnification, indicates that the image formed is virtual, erect, and diminished.	
Section –E		
Question No.37 to 39 are case-based/data –based questions.		
37	A. The gas released is carbon dioxide (CO₂). (1) B. The balanced chemical equation is: $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$ (2) OR C. A decomposition reaction is a reaction in which a compound breaks down into two or more simpler substances. The general form of a decomposition reaction is: AB → A+B. D. Carbon dioxide reacts with lime water (Ca(OH)₂) to form calcium carbonate (CaCO₃), which makes the lime water turn milky. (1)	4
38	A. To prevent it from receiving sunlight, thus inhibiting photosynthesis in that plant. (2) OR B. No starch production, indicating that it could not perform photosynthesis, while the uncovered plant will show starch production.	4

	C. Absorbs light energy, which is necessary for converting carbon dioxide and water into glucose during photosynthesis. (1) D. By showing that the plant that received no light (the covered plant) could not produce starch, while the plant exposed to light was able to perform photosynthesis effectively. (1)	
39	A. $R_s = 4 + 6 = 10 \Omega$ (1) B. $I = V/R = 24/10 = 2.4 \text{ A}$ (1) C. $V = I \times R = 2.4 \times 4 = 9.6 \text{ V}$ (2) OR D. $1/R_p = 1/4 + 1/6 = 5/12$, $R_p = 2.4 \Omega$ Total current $= 24/2.4 = 10 \text{ A}$	4